

SUMMARY

- A network function is defined as the ratio of the zero-state response transform to the input transform. Network functions are either driving-point functions or transfer functions. Network functions are rational functions of s with real coefficients whose complex poles and zeros occur in conjugate pairs.
- Network functions for simple circuits like voltage and current dividers and inverting and noninverting OP AMPs are easy to derive and often useful. Node-voltage or mesh-current methods are used to find the network functions for more complicated circuits. The transfer function of a cascade connection obeys the chain rule when each stage does not load the preceding stage in the cascade.
- The impulse response is the zero-state response of a circuit for a unit impulse input. The transform of the impulse response is equal to the network function. The impulse response contains only natural poles and decays to zero in stable circuits. The impulse response of a linear, time-invariant circuit obeys the proportionality and time-shifting properties. The short pulse approximation is a useful way to simulate the impulse response in practical situations.
- The step response is the zero-state response of a circuit when the input is a unit step function. The transform of the step response is equal to the network function times $1/s$. The step response contains natural poles and a forced pole at $s = 0$ that leads to a dc steady-state response in stable circuits. The amplitude of the dc steady-state response can be found by evaluating the network function at $s = 0$. The step response waveform can also be found by integrating the impulse response waveform.
- The sinusoidal steady-state response is the forced response of a stable circuit for a sinusoidal input. With a sinusoidal input the response transform contains natural poles and forced poles at $s = \pm j\omega$ that lead to a sinusoidal steady-state response in stable circuits. The amplitude and phase angle of the sinusoidal steady-state response can be found by evaluating the network function at $s = j\omega$.
- The sinusoidal steady-state response can be found using phasor circuit analysis or directly from the transfer function. Phasor circuit analysis works best when the circuit is driven at only one frequency and several responses are needed. The transfer function method works best when the circuit is driven at several frequencies and only one response is needed.
- The convolution integral is a t -domain method relating the impulse response $h(t)$ and input waveform $x(t)$ to the zero-state response $y(t)$. Symbolically the convolution integral is represented by $y(t) = h(t) * x(t)$. Time-domain convolution and s -domain multiplication are equivalent; that is, $y(t) = h(t) * x(t) = \mathcal{L}^{-1}\{H(s)X(s)\}$. The geometric interpretation of t -domain convolution involves four operations: reflecting, shifting, multiplying, and integrating.
- First- and second-order transfer functions can be designed using voltage dividers and inverting or noninverting OP AMP circuits. Higher-order transfer functions can be realized using a cascade connection of first- and second-order circuits. Prototype designs usually require magnitude scaling to obtain practical element values.

PROBLEMS

OBJECTIVE 11-1 NETWORK FUNCTIONS
(SECTS. 11-1, 11-2)

Given a linear circuit:

- Find specified network functions and locate their poles and zeros.
- Select the element values to produce specified poles and zeros.

See Examples 11-1, 11-2, 11-3, 11-4, 11-5, 11-6, 11-7 and Exercises 11-1, 11-2, 11-3, 11-4, 11-5, 11-6, 11-7, 11-8.

- 11-1 Find the driving point impedance seen by the voltage source in Figure P11-1 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

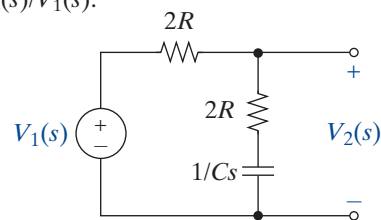


FIGURE P11-1

- 11-2 Find the driving point impedance seen by the voltage source in Figure P11-2 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

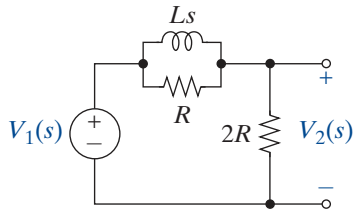


FIGURE P11-2

- 11-3 Find the driving point impedance seen by the voltage source in Figure P11-3 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

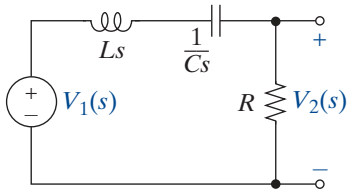


FIGURE P11-3

- 11-4 Find the driving point impedance seen by the voltage source in Figure P11-4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

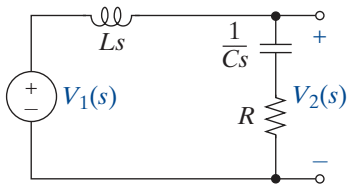


FIGURE P11-4

- 11-5 Find the driving point impedance seen by the voltage source in Figure P11-5 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

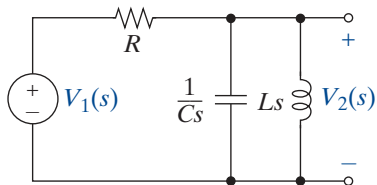


FIGURE P11-5

- 11-6 Find the driving point impedance seen by the voltage source in Figure P11-6 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

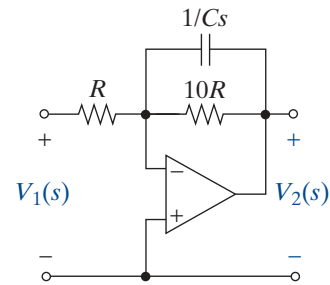


FIGURE P11-6

- 11-7 Find the driving point impedance seen by the voltage source in Figure P11-7 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

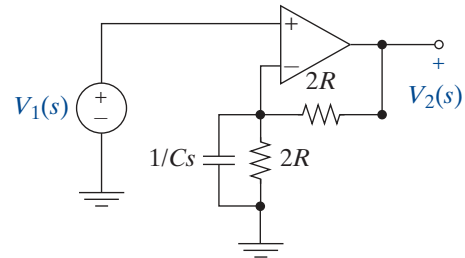


FIGURE P11-7

- 11-8 Find the driving point impedance seen by the voltage source in Figure P11-8 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

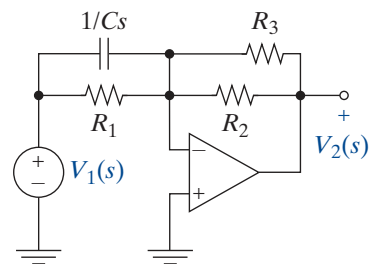


FIGURE P11-8

- 11-9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-9.

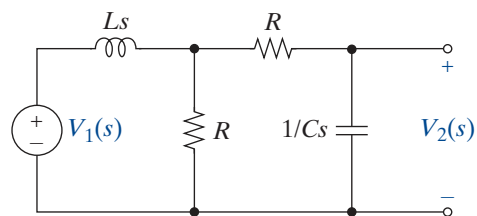


FIGURE P11-9

- 11–10 Find the driving point impedance seen by the voltage source in Figure P11–10 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$. Insert a follower at A and repeat.

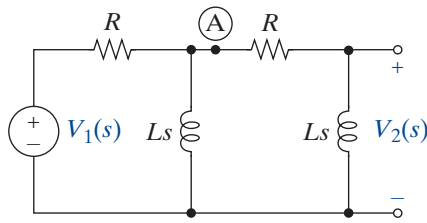


FIGURE P11–10

- 11–11 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11–11. Select values of R and C so that $T_V(s)$ has a pole at $s = -100$ krad/s.

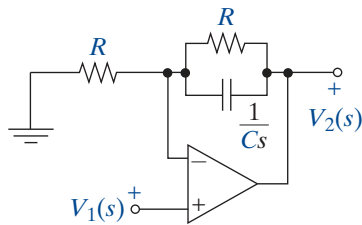


FIGURE P11–11

- 11–12 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11–12. Select values of R_1 , R_2 , and C so that $T_V(s)$ has a pole at $s = -250$ krad/s and $R_2/R_1 = 100$.

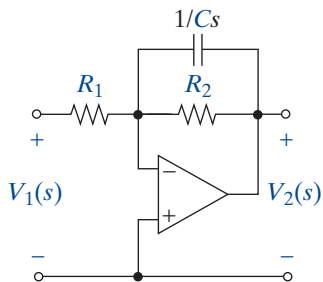


FIGURE P11–12

- 11–13 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11–13. Select values of R and L so that $T_I(s)$ has a pole at $s = -377$ rad/s.

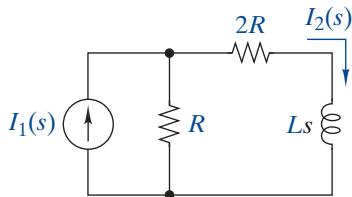


FIGURE P11–13

- 11–14 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–14. Locate the poles and zeros of the transfer function.

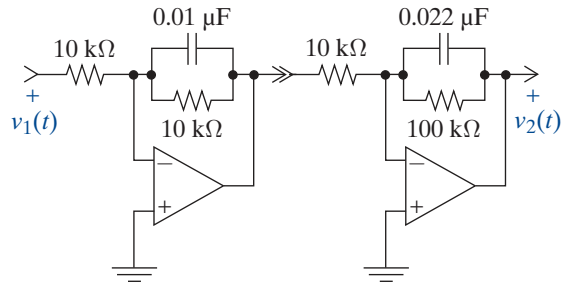


FIGURE P11–14

- 11–15 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–15. Locate the poles and zeros of the transfer function.

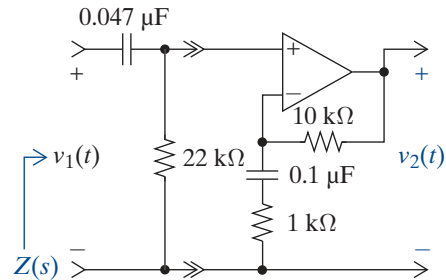


FIGURE P11–15

- 11–16 Find the input impedance $Z(s)$ in Figure P11–15.

OBJECTIVE 11–2 NETWORK FUNCTIONS, IMPULSE RESPONSE, AND STEP RESPONSES (SECTS. 11–3, 11–4)

- Given a first- or second-order linear circuit, find its impulse or step response.
- Given the impulse or step response of a linear circuit, find the network functions.
- Given the impulse or step response of a linear circuit, find the response due to other inputs.

See Examples 11–8, 11–9, 11–10, 11–11, 11–12 and Exercises 11–9, 11–10, 11–11, 11–12, 11–13, 11–14.

- 11–17 Find the impulse response at $v_2(t)$ in Figure P11–17. Find the circuit's step response.

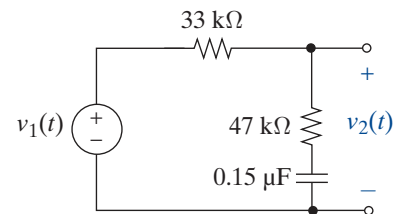


FIGURE P11–17

- 11–18 Find $v_2(t)$ in Figure P11–18 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

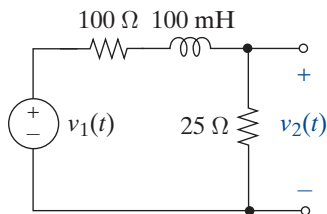


FIGURE P11–18

- 11–19 Find $v_2(t)$ in Figure P11–19 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

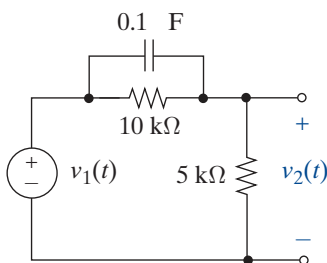


FIGURE P11–19

- 11–20 Find $h(t)$ and $g(t)$ for the circuit in Figure P11–20.

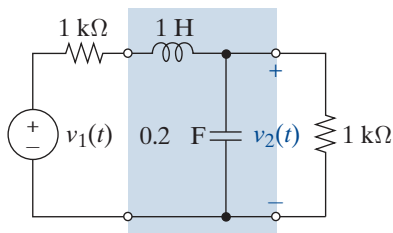


FIGURE P11–20

- 11–21 Find $h(t)$ and $g(t)$ for the circuit in Figure P11–21 if $R_F = 100 \text{ k}\Omega$.

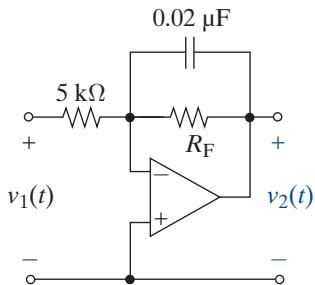


FIGURE P11–21

- 11–22  Select an appropriate R_F for the circuit of Figure P11–21 so that the step response of the circuit is $g(t) = (10e^{-1000t} - 10) u(t) \text{ V}$.

- 11–23 Find $v_2(t)$ in Figure P11–23 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

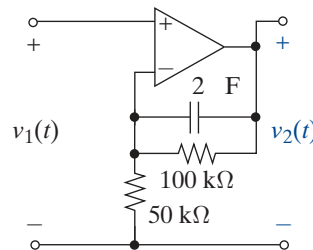


FIGURE P11–23

- 11–24 The impulse response of a linear circuit is $h(t) = (100e^{-200t} - 100e^{-1000t})\mu(t)$. Find the circuit's step response $g(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11–25 The impulse response of a linear circuit is $h(t) = \delta(t) - 2000e^{-200t} u(t)$. Find the circuit's step response $g(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11–26 The step response transform of a linear circuit is $G(s) = 1000/(s + 1000)$. Find the circuit's impulse response $h(t)$, step response $g(t)$, impulse response transform $H(s)$, and the circuit's transfer function $T(s)$.

- 11–27 The step response of a linear circuit is $g(t) = 15(e^{-20kt} - e^{-30kt})u(t)$. Find the circuit's impulse response $h(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11–28 Find $h(t) = \frac{dg(t)}{dt}$ when $g(t) = (3 - e^{-10t}) u(t)$. Verify your answer by first transforming $g(t)$ into $G(s)$ and finding $H(s) = sG(s)$ and then taking the inverse transform of $H(s)$. Did you get the same answer?

- 11–29 The impulse response of a linear circuit is $h(t) = 1000 [e^{-1000t}]u(t)$. Find the output waveform when the input is $x(t) = 5tu(t)$.

- 11–30  The step response of a linear circuit is $g(t) = 0.25[1 - e^{-150t}]u(t)$. Find the output waveform when the input is $v_1(t) = [20e^{-200t}]u(t)$. Use MATLAB to find the Laplace transforms of $g(t)$ and $v_1(t)$. Then find $V_2(s)$. Finally, use the inverse Laplace function to find the waveform $v_2(t)$ and plot the results.

- 11–31 The step response of a linear circuit is $g(t) = 10[e^{-50t} \cos 200t]u(t)$. Find the circuit's impulse response $h(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11–32  The impulse response transform of a linear circuit is $H(s) = (s + 2000)/(s + 1000)$. Find the output waveform when the input is $x(t) = 5e^{-1000t}u(t)$. Use MATLAB to find the Laplace transform of $x(t)$. Then find $Y(s)$. Finally, use the inverse

Laplace function to find the waveform $y(f)$ and plot the results.

- 11–33 The impulse response of a linear circuit is $h(t) = 20u(t) + \delta(t)$. Find the output waveform $y(t)$ when the input is $x(t) = 2[e^{-20t}]u(t)$.

OBJECTIVE 11–3 NETWORK FUNCTIONS AND THE SINUSOIDAL STEADY-STATE RESPONSE (SECT. 11–5)

- (a) Given a first- or second-order linear circuit with a specified input sinusoid, find the sinusoidal steady-state response.
 (b) Given the network function, impulse response, or step response, find the sinusoidal steady-state response for a specified input sinusoid.

See Examples 11–13, 11–14, 11–15 and Exercises 11–15, 11–16, 11–17, 11–18.

- 11–34 The circuit in Figure P11–34 is in the steady state with $v_1(t) = 5 \cos 1414.21t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 5 \cos 1 \text{ kt}$ V. And without doing any calculations, repeat for $v_1(t) = 5$ V.

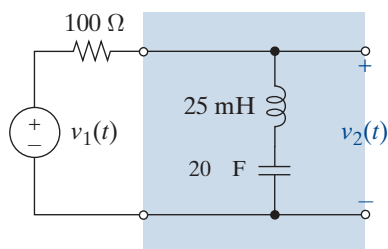


FIGURE P11–34

- 11–35 The circuit in Figure P11–35 is in the steady state with $v_1(t) = 10 \cos 500t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 10 \cos 1 \text{ kt}$ V, and for $v_1(t) = 10 \cos 10 \text{ kt}$ V. Where is the pole located?

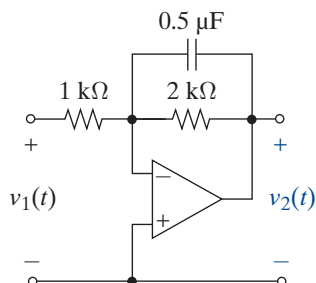


FIGURE P11–35

- 11–36 The circuit in Figure P11–36 is in the steady state with $v_1(t) = 25 \cos 2000t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 25 \cos 10 \text{ kt}$ V. Where are the poles located?

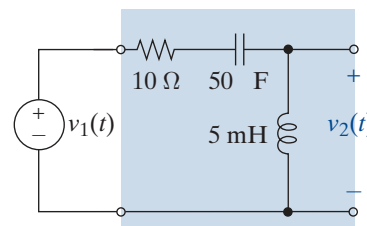


FIGURE P11–36

- 11–37 The circuit in Figure P11–37 is in the steady state with $i_1(t) = 10 \cos 50kt$ mA, $R_1 = 100 \Omega$, $R_2 = 400 \Omega$, and $L = 100 \text{ mH}$. Find $i_{2ss}(t)$. Repeat for $i_1(t) = 10 \cos 5kt$ mA. Where is the pole located?

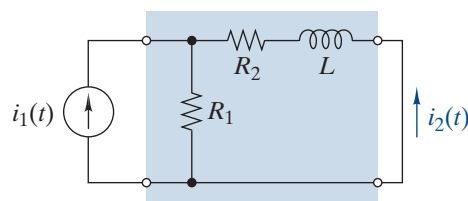


FIGURE P11–37

- 11–38 The circuit in Figure P11–38 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 0.5 \text{ }\mu\text{F}$.
 (a) Find $i_{2ss}(t)$.

(b) Verify your results using OrCAD. Use IAC for a source and note that the current comes out of the negative terminal (passive sign convention.). The IPRINT element on the right side of the figure is an ammeter and it comes from the Special Library. It has a polarity as indicated by the small “–” on the element. You should ensure that it will read the correct current direction. Before you can use the IPRINT element, it needs to be set up to display the current magnitude and phase. Double click on the element and in its property editor place a “y” in AC, MAG, and PHASE boxes. Click the Apply button and close the property editor. To obtain the simulation select AC Sweep/Noise. Set the start frequency and the stop frequency at 159.15 Hz (1000 rad/s). Place a 1 in Points/Decade. Run the simulation. Under View, select Output File. Scroll down until you see the IPRINT output.

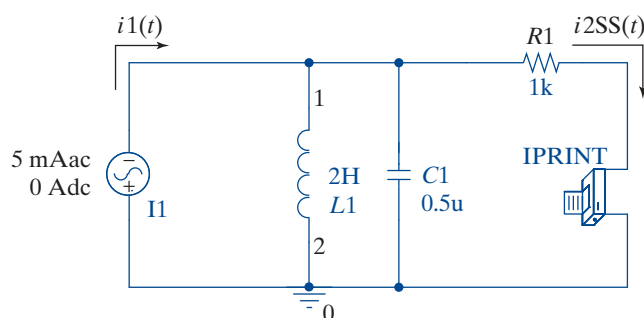


FIGURE P11–38

2 mHz to 2 kHz must fall in the transducer passband, which means that the cutoff frequency must be less than 2 mHz. In equation form this requires

$$\omega_C = \frac{1}{T_C} < 2\pi \times 0.002 \quad \text{or} \quad T_C > \frac{250}{\pi} = 79.6 \text{ s}$$

Models 266, 268, and 269 meet the frequency-response requirement, since they have discharge time constants T_C greater than 79.6 s. Model 266 is the best choice because it has the most sensitivity (passband gain) at $K_V = 5 \text{ mV/psi}$. ■

SUMMARY

- The frequency response of a circuit is defined by the variation of the gain $|T(j\omega)|$ and phase $\angle T(j\omega)$ with frequency. The gain function is usually expressed in dB in frequency-response plots. Logarithmic frequency scales are used on frequency-response plots of the gain and phase functions.
- A passband is a range of frequencies over which the steady-state output is essentially constant with very little attenuation. A stopband is a range of frequencies over which the steady-state output is significantly attenuated. The cutoff frequency is the boundary between a passband and the adjacent stopband.
- Circuit gain responses are classified as low pass, high pass, bandpass, and bandstop depending on the number and location of the stopbands and passbands. The performance of devices and circuits is often specified in terms of frequency-response descriptors such as bandwidth, passband gain, and cutoff frequency.
- The low- and high-frequency gain asymptotes of a first-order circuit intersect at a corner frequency determined by the location of its pole. The total phase change from low to high frequency is $\pm 90^\circ$. First-order circuits can be connected to produce bandpass and bandstop responses.
- Series and parallel RLC circuits provide bandpass and bandstop gain characteristics that are easily related to the circuit parameters.
- Bode plots are graphs of the gain (in dB) and phase angle (in degrees) versus log-frequency scales. Straight-line approximations to the gain and phase can be constructed using the corner frequencies defined by the poles and zeros of $T(s)$. The purpose of the straight-line approximations is to develop a conceptual understanding of frequency response. The straight-line plots do not necessarily provide accurate data at all frequencies, especially for circuits with complex poles.
- Computer-aided circuit analysis programs can accurately generate and plot frequency-response data. The user must have a rough idea of the gain and frequency ranges of interest to use these tools intelligently.
- It is often necessary to relate the time-domain characteristics of a circuit to its frequency response, and vice versa. The transfer function provides a link between the frequency-domain and time-domain responses.

PROBLEMS

OBJECTIVE 12-1 FIRST-ORDER CIRCUIT FREQUENCY RESPONSE (SECTS. 12-1, 12-2, 12-3)

Given a first-order circuit or transfer function:

- Find and classify the frequency response.
- Plot the gain and phase responses using straight-line approximations and computer tools.
- Design circuits to produce a specified frequency response. See Examples 12-2, 12-3, 12-4, 12-5, 12-6, 12-7, 12-8 and Exercises 12-3, 12-4, 12-5, 12-6, 12-7, 12-8, 12-9, 12-10, 12-11, 12-12, 12-13.

12-1 A transfer function has a passband gain of 500. At a particular frequency in its stopband, the gain of the transfer

function is only 0.00025. By how many decibels does the gain of the passband exceed that of that frequency in the stopband?

12-2 A particular filter is said to be 56 dB down at a desired stop frequency. How many times reduced is a signal at that frequency compared to a signal in the filter's pass-band?

12-3 A certain low-pass filter has the Bode diagram shown in Figure P12-3.

(a) How many dB down is the filter at 5000 rad/s?

(b) Estimate where the cutoff frequency occurs, then determine how many dB down is the filter at one decade after the cutoff frequency?

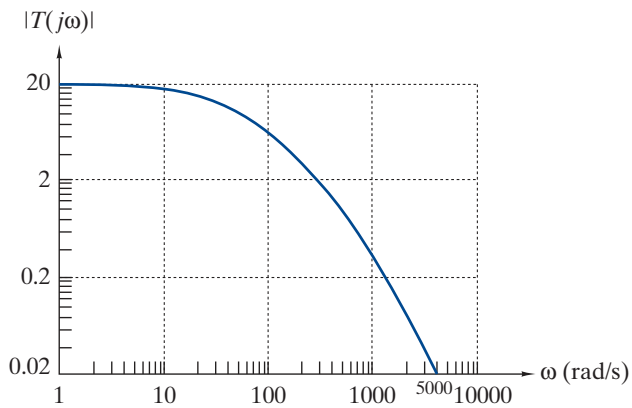


FIGURE P12-3

12-4 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-4.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximations of the gain and phase responses.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- How many dB down from the passband is the filter at one octave past the cutoff?

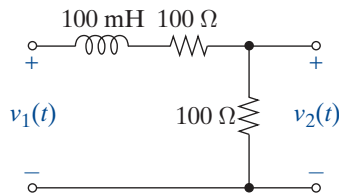


FIGURE P12-4

12-5 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-5.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- How many dB down from the passband is the filter at one octave before the cutoff?

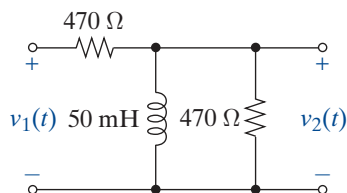


FIGURE P12-5

12-6 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-6.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.1\omega_C$, ω_C , and $10\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- What element values would you change to increase the passband gain to 10?
- How many dB down from the passband is the filter at one decade past the cutoff?

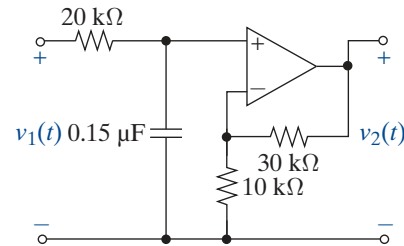


FIGURE P12-6

12-7 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-7.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.1\omega_C$, ω_C , and $10\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- What element values would you change to double the passband gain without changing the cutoff frequency?

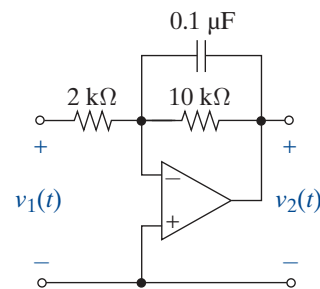


FIGURE P12-7

12-8 Your task is to connect the modules in Figure P12-8 so that the gain of the transfer function is 4 and the cutoff frequency of the filter is 500 rad/s when connected between the source and the load. Repeat if the cutoff frequency is 625 rad/s and the gain did not matter.

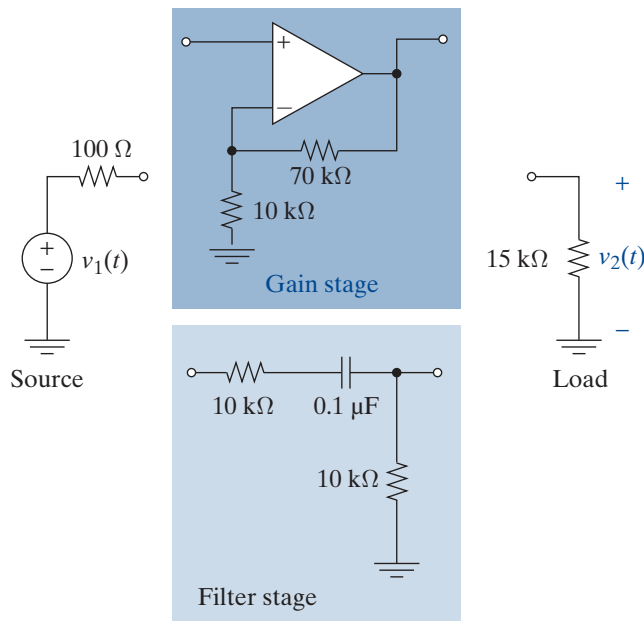


FIGURE P12-8

- 12-9 **E** A young designer needed to design a low-pass filter with a cutoff of 1 krad/s and a gain of -5 . The filter is to fit as an interface between the source and the load. The designer was perplexed when no matter how the stages shown in Figure P12-9 are connected the results are not what were expected. Explain the problem and suggest a way to achieve the desired results.

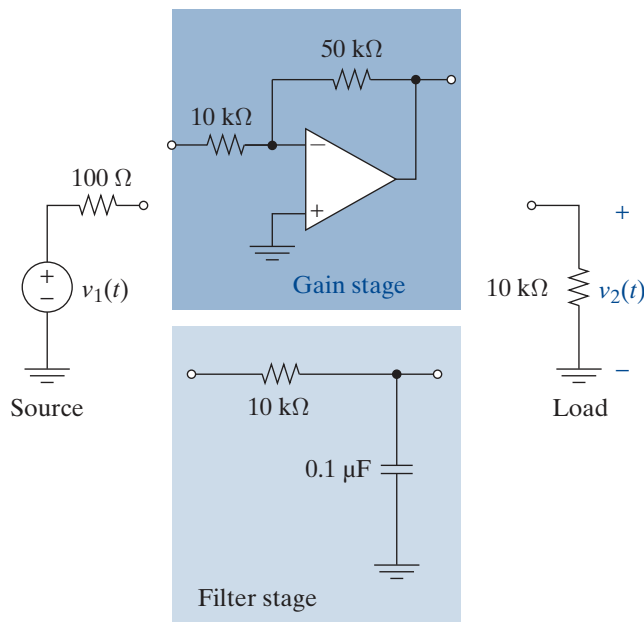


FIGURE P12-9

- 12-10 **D** Design a low-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of 1.

- 12-11 **D** Design an RC low-pass first-order filter with a cutoff frequency of 10,000 rad/s and a passband gain of $+10$.

- 12-12 **D** Design an RL high-pass first-order filter with a cutoff frequency of 120 Hz and a passband gain of -15 .

- 12-13 **D** Design an RC high-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of -100 .

- 12-14 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-14.

- (a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

- (b) Use OrCAD to plot the Bode magnitude gain response of the circuit.

- (c) What element value would you change to increase the cutoff frequency by one decade?

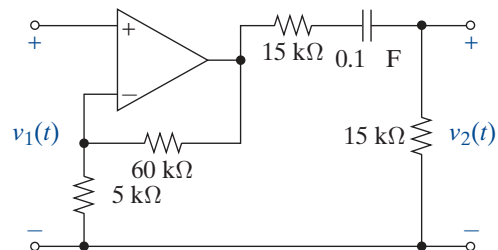


FIGURE P12-14

- 12-15 **D** Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-15. What type of gain response does the circuit have? What is the passband gain? Select practical, standard values of R and C from the inside rear cover so that the cutoff frequency is 300 kHz $\pm 5\%$.

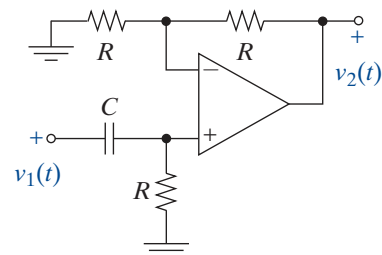


FIGURE P12-15

- 12-16 A first-order high-pass circuit has a passband gain of 20 dB and a cutoff frequency of 1000 rad/s. Find the gain (in dB) at $\omega = 0, 500, 1000$, and 5000 rad/s.

- 12-17 A first-order low-pass circuit has a passband gain of 0 dB and a cutoff frequency of 2 krad/s. Find the gain (in dB) at $\omega = 0, 200$ rad/s, 20 krad/s, and 200 krad/s.

- 12-18 **D** The transfer function of a first-order circuit is

$$T(s) = \frac{10,000}{s + 2000}$$

- (a) Identify the type of gain response. Find the cutoff frequency and the passband gain.

- (b) Use MATLAB to plot the magnitude of the Bode gain response.

- (c) Design a circuit to realize the transfer function.
 (d) Use OrCAD to validate your circuit design by comparing its frequency response to the MATLAB output.

12-19 **D** The transfer function of a first-order circuit is

$$T(s) = \frac{5s}{s + 15000}$$

- (a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
 (b) Use MATLAB to plot the magnitude of the Bode gain response.
 (c) Design a circuit to realize the transfer function.
 (d) Use OrCAD to validate your circuit design by comparing its frequency response to the MATLAB output

12-20 **D** A student decided that she needed a low-pass filter that had a roll-off of -2 or -40dB/decade , with a cutoff frequency of 1000 rad/s . She correctly designed two identical passive RC filters, each with a cutoff of 1000 rad/s , and connected them in cascade or one after the other. She cleverly separated them with a buffer to avoid loading. When she measured the cutoff frequency she was dismayed to discover that it was not 1000 rad/s although the roll-off was correct.
 (a) Simulate the cascaded pair and determine where the actual cutoff frequency occurs.
 (b) Help the student by altering her design so that the two identical RC filters actually produce the desired outcome. Explain why the discrepancy occurred.

OBJECTIVE 12-2 BANDPASS AND BANDSTOP RESPONSES (SECT. 12-4)

Given a cascade or parallel connection of two first-order circuits:

- (a) Find and classify the frequency response.
 (b) Plot the gain and phase responses using straight-line approximations and computer tools.
 (c) Design circuits to produce a specified frequency response. See Examples 12-9, 12-10 and Exercises 12-14, 12-15, 12-16.

12-21 **D** The circuit in Figure P12-21 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 100 and the cutoff frequencies are 1000 rad/s and 40 krad/s . Use practical element values with $R \geq 10\text{ k}\Omega$ and $C \leq 1\text{ }\mu\text{F}$. Use OrCAD to validate your design.

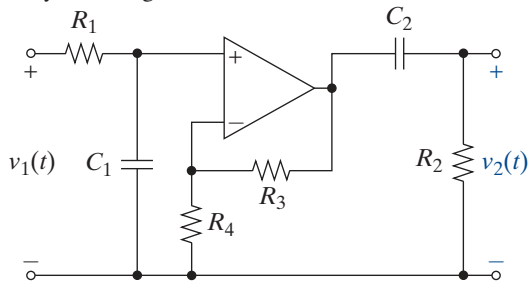


FIGURE P12-21

12-22 **D** Repeat Problem 12-21 for the circuit in Figure P12-22. Use OrCAD to validate your design.

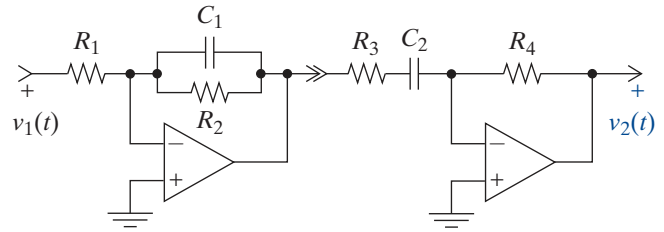


FIGURE P12-22

12-23 **E** Suppose the circuits you designed in problems 12-21 and 12-22 had to feed a $500\text{ }\Omega$ instrument. Which circuit would you select and why?

12-24 **D** The circuit in Figure P12-24 produces a bandstop response for a suitable choice of element values.

- (a) Find the circuit's transfer function.
 (b) Identify the elements that control the two cutoff frequencies. Select the element values so that the cutoff frequencies are 200 krad/s and 4000 krad/s . Use practical element values with $R_s \geq 10\text{ k}\Omega$ and $C_s \leq 1\text{ }\mu\text{F}$. Design your passband gain to be $+20\text{ dB}$.
 (c) Use MATLAB to plot the Bode magnitude plot for the values you selected.
 (d) Simulate your circuit using OrCAD and compare the results to the MATLAB output.

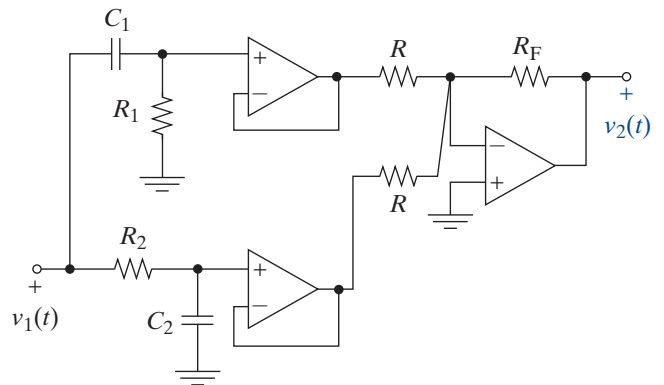


FIGURE P12-24

12-25 **D** Design an audio amplifier that amplifies signals from 20 Hz to 20 kHz . Your approach should be to use a cascade connection of two first-order passive circuits separated by a non-inverting OP AMP. The source has a $50\text{-}\Omega$ series resistor and the output of the filter feeds a $10\text{-k}\Omega$ audio transducer. Design a bandpass circuit with the following specifications all $\pm 5\%$: $f_{CL} = 20\text{ Hz}$, $f_{CH} = 20\text{ kHz}$, $B = 19980\text{ Hz}$. Passband gain $+40\text{ dB}$. Use OrCAD to validate your results.

12-26 **D-E** Design an audio amplifier that amplifies signals from 2 kHz to 10 kHz . Your approach should be to use a

cascade connection of two first-order passive circuits separated by a non-inverting OP AMP. The source has a $50\text{-}\Omega$ series resistor and the output of the filter feeds a $10\text{-k}\Omega$ audio transducer. Design a bandpass circuit with the following specifications all $\pm 5\%$: $f_{CL} = 2\text{ kHz}$, $f_{CH} = 10\text{ kHz}$, $B = 8\text{ kHz}$. Passband gain $+40\text{ dB}$. Use OrCAD to validate your results. Even though problems 12–25 and 12–26 share similar constructs one cannot achieve the desired specifications in 12–26. What is the reason for the problem?

12–27 Design an audio amplifier that amplifies signals from 20 Hz to 20 kHz . Your approach should be to use a cascade connection of two first-order active OP AMP circuits. The source has a $50\text{-}\Omega$ series resistor and the output of the filter feeds a $10\text{-k}\Omega$ audio transducer. Design a bandpass circuit with the following specifications all $\pm 5\%$: $f_{CL} = 20\text{ Hz}$, $f_{CH} = 20\text{ kHz}$, $B = 19980\text{ Hz}$. Passband gain should be adjustable (use a potentiometer) from 0 dB to $+40\text{ dB}$. Use OrCAD to validate your results.

12–28 A student needed to design a bandstop filter that was to block frequencies between 1000 rad/s and 10000 rad/s with unity gain in the passbands. His design is shown in Figure P12–28. As a teaching assistant, you are required to grade his design. What grade would you assign and what critique would you give him?

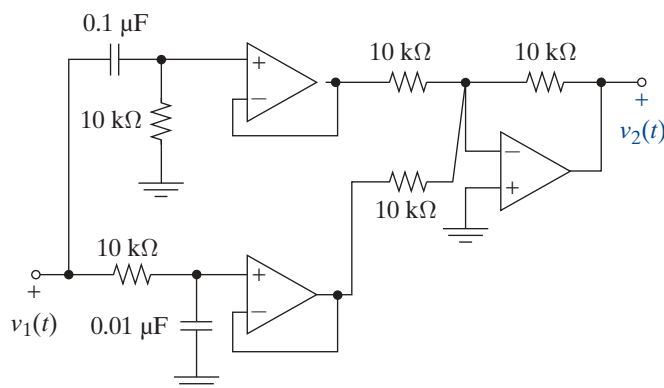


FIGURE P12–28

OBJECTIVE 12–3 THE FREQUENCY RESPONSE OF *RLC* CIRCUITS (SECT. 12–5)

Given an *RLC* circuit connected as a bandpass or a bandstop filter:

- Find the frequency-response descriptors such as Q and B .
- Design circuits to produce a specified frequency response.
- Compare the bandpass and bandstop responses obtained from *RLC* filters with similar ones obtained from first-order filters.

See Examples 12–11, 12–12, 12–13, 12–14, 12–15 and Exercises 12–17, 12–18, 12–19, 12–20, 12–21

12–29 Determine the filter type for the circuit in Figure P12–29. Then find Q , B , ω_{C1} , ω_{C2} , and ω_0 . Is the circuit a narrow-band or a wide-band filter?

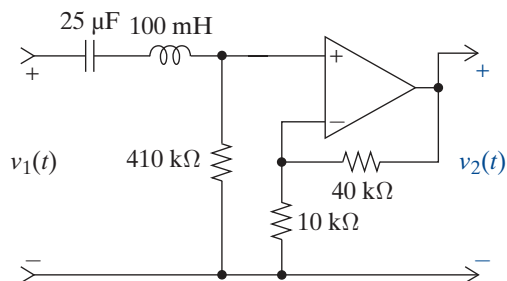


FIGURE P12–29

12–30 Design an *RLC* bandstop filter with a center frequency of 33 krad/s and a bandwidth of 3.3 krad/s . The passband gain is 0 dB . Use practical values for R , L , and C .

12–31 Design an *RLC* bandpass filter with a center frequency of 1000 rad/s and a Q of 20 . The passband gain is $+20\text{ dB}$. Use practical values for R , L , and C .

12–32 A series *RLC* bandpass circuit with $R = 30\text{ }\Omega$ is designed to have a bandwidth of 5 Mrad/s and a center frequency of 50 Mrad/s . Find L , C , Q , and the two cutoff frequencies. Could you design this circuit using a cascade connection of two first-order filters separated by a follower? Why or why not?

12–33 A parallel *RLC* bandpass circuit with $C = 0.01\text{ }\mu\text{F}$ and $Q = 10$ has a center frequency of 500 krad/s . Find R , L , and the two cutoff frequencies. Could you design this circuit using a cascade connection of two first-order filters separated by a follower? Why or why not?

12–34 A 20-mH inductor with an internal series resistance of $25\text{ }\Omega$ is connected in series with a capacitor and a voltage source with a Thévenin resistance of $50\text{ }\Omega$.

- What value of C is needed to produce $\omega_0 = 5\text{ krad/s}$?
- Find the bandwidth and quality factor of the circuit.

12–35 In a series *RLC* circuit, which element would you adjust (and by how much) to

- Double the bandwidth without changing the center frequency?
- Double the center frequency without changing the bandwidth?
- Repeat parts (a) and (b) for a parallel *RLC* circuit.

12–36 A parallel *RLC* circuit with $R = 1\text{ k}\Omega$ has a center frequency of 50 kHz and a bandwidth of 50 kHz . Find the values of L and C . Could you design this circuit using a cascade connection of two first-order filters separated by a follower? Why or why not?

12–37 A series *RLC* bandpass filter is required to have resonance at $f_0 = 50\text{ kHz}$. The circuit is driven by a sinusoidal source with a Thévenin resistance of $60\text{ }\Omega$. The following standard capacitors are available in the stock room: $1\text{ }\mu\text{F}$, $0.68\text{ }\mu\text{F}$, $0.47\text{ }\mu\text{F}$, $0.33\text{ }\mu\text{F}$, $0.2\text{ }\mu\text{F}$, and $0.12\text{ }\mu\text{F}$. The inductor will be custom-designed to match the capacitor used. Select the available capacitor that minimizes the filter bandwidth.

12-38 A series RLC bandstop circuit is to be used as a notch filter to eliminate a bothersome 120-Hz hum in an audio channel. The signal source has a Thévenin resistance of $300\ \Omega$. Select values of L and C so the upper cutoff frequency of the stopband is below 180 Hz. Use OrCAD to verify your design.

12-39 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ for the bandpass circuit in Figure P12-39. Use MATLAB to visualize the Bode characteristics if $R = 50\ \Omega$, $L = 500\ \mu\text{H}$, and $C = 0.002\ \mu\text{F}$. Design an active circuit to meet those characteristics. Verify your design using OrCAD.

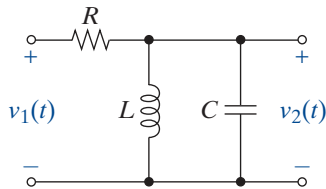


FIGURE P12-39

12-40 Show that the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-40 has a bandstop filter characteristic. Derive expressions relating the notch frequency and the cutoff frequencies to R , L , and C . Then select values of R , L , and C so that the bandwidth and the center frequency are both 10 krad/s. Validate your design using OrCAD.

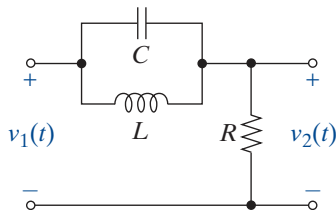


FIGURE P12-40

12-41 Figure P12-41 shows an RLC filter with an input current and an output voltage. The purpose of this problem is to determine the filter type using informal circuit analysis. Use the element impedances and basic analysis tools to find the magnitude of the output voltage $|V_2(j\omega)|$ at $\omega = 0$, $\omega = \infty$, and $\omega = \omega_0 = 1/\sqrt{LC}$. What is the filter type?

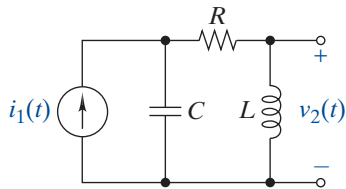


FIGURE P12-41

12-42 A student was designing a passive tuned filter using a series RLC circuit with $R = 10\ \Omega$, $L = 10\ \mu\text{H}$, and $C =$

$0.025\ \mu\text{F}$. When he connected the output of the filter across the capacitor he was surprised by an output near the cutoff point that was greater than the input signal. Since the filter was passive he was quite perplexed. Explain how this can be.

OBJECTIVE 12-4 BODE PLOTS (SECTS. 12-6, 12-7)

Given a linear circuit or transfer function:

- Plot the gain and phase responses using straight-line approximations and computer tools.
- Develop a transfer function from a straight-line Bode gain plot.
- Design a circuit that produces a given straight-line gain plot. See Examples 12-16, 12-17, 12-18, 12-19 and Exercises 12-22, 12-23, 12-24, 12-25, 12-26.

12-43 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ for the circuit in Figure P12-43.

(a) Construct the straight-line Bode plot of the gain and phase of the transfer function. Use the straight-line plots to estimate the amplitude and phase of the steady-state output for $v_1(t) = 10 \sin 100t\ \text{V}$.

(b) Calculate the actual output amplitude and phase for this input and compare the two results.

(c) Use MATLAB to generate a Bode plot of your transfer function. Compare your answers from the straight-line plot and your hand calculations, with that produced by MATLAB.

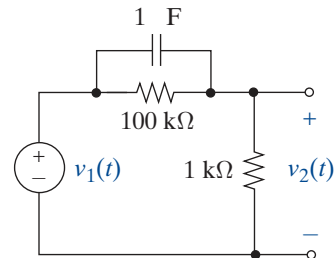


FIGURE P12-43

12-44 Repeat Prob. 12-43 using the circuit in Figure P12-44.

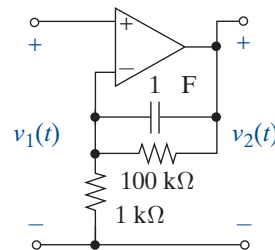


FIGURE P12-44

12-45 For the following transfer function

$$T(s) = \frac{10(s + 100)}{(s + 1)}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.
- (b) Use MATLAB to plot the Bode magnitude of the transfer function.
- (c) Design a circuit using practical values for the components that has the same transfer function.
- (d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-46 **D** For the following transfer function

$$T(s) = \frac{100(s+10)}{s(s+100)}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.
- (b) Use MATLAB to plot the Bode magnitude of the transfer function.
- (c) Design a circuit using practical values for the components that has the same transfer function.
- (d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-47 **D** For the following transfer function

$$T(s) = \frac{10^6 s^2}{(s+1000)^2}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.
- (b) Use MATLAB to plot the Bode magnitude of the transfer function.
- (c) Design a circuit using practical values for the components that has the same transfer function.
- (d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-48 **D** For the following transfer function

$$T(s) = \frac{100s(s+300)}{(s+30)(s+3000)}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.
- (b) Use MATLAB to plot the Bode magnitude of the transfer function.
- (c) Design a circuit using practical values for the components that has the same transfer function.
- (d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-49 **D** For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T_V(s) = \frac{10(s+10)(s+100)}{(s+1)(s+1000)}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency(ies) and passband gain.
- (b) Use MATLAB to plot the Bode magnitude and phase of the transfer function.
- (c) What is the output $v_2(t)$, if $v_1(t) = 10 \cos(200t + 45^\circ)$ V?

12-50 **D** For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T(s) = \frac{10^8(s+100)^2}{(s+1000)^4}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency(ies) and passband gain.
- (b) Use MATLAB to plot the Bode magnitude and phase of the transfer function.
- (c) What is the output $v_2(t)$, if $v_1(t) = 10 \cos(1000t - 60^\circ)$ V?

12-51 **D** For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T_V(s) = \frac{4s}{0.04s^2 + 0.2s + 1}$$

- (a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function?
- (b) Use the straight-line plot to estimate the maximum gain and the frequency at which it occurs.
- (c) Use MATLAB to plot the Bode magnitude and phase of the transfer function.

12-52 **D** Consider the gain plot in Figure P12-52.

- (a) Find the transfer function corresponding to the straight-line gain plot.
- (b) Use MATLAB to plot the Bode magnitude of the transfer function.
- (c) Compare the straight-line gain and the actual gain at $\omega = 20$ and 200 rad/s.
- (d) Design a circuit to realize the Bode plot.

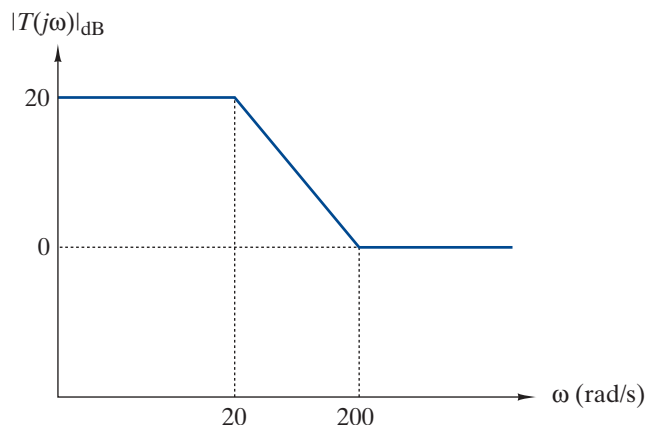


FIGURE P12-52